

COVER PAGE

BUSINESS REQUIREMENTS DOCUMENT
EXPLORING SPACE, THE MILKY WAY GALAXY AND
THE UNIVERSE

A Comprehensive Scientific Requirements, Knowledge and Exploration Framework

Document Field	Value
Prepared By	Research & Business Analysis Team
Organization / Team	Space Science Knowledge Initiative
Version	1.0
Date	15 September 2026
Document Status	Baseline Draft for Review
Primary Use	Academic, scientific, research and professional documentation

Purpose Business Requirements Document for a structured scientific and educational framework covering the Universe, the observable cosmos, the Milky Way Galaxy, the Solar System, stellar and galactic phenomena, modern astronomy, space exploration, and future research priorities.

This document treats astronomy as a requirements-driven knowledge and research domain. It identifies the scientific scope, evidence base, observation needs, analytical requirements, technology dependencies, uncertainties, risks, and acceptance criteria that would govern a serious space-science program or educational research initiative.

Prepared for: Academic, scientific, research, portfolio and professional documentation use. Prepared by: Research & Business Analysis Team. Organization/Team: Space Science Knowledge Initiative. Version: 1.0. Date: 15 September 2026. Status: Baseline Draft for Review.

Usage statement: This document is intended for scientific education, research planning, requirements analysis and professional documentation. It is not a substitute for peer-reviewed literature, mission operations documentation, engineering certification, or original astronomical data. Claims are framed according to established scientific understanding unless explicitly identified as theoretical, model-dependent or unresolved.

Scope statement The document progresses from cosmic origin and structure through the Milky Way, stars, planets, exploration systems, scientific data and future research, with explicit treatment of evidence, uncertainty, risk and acceptance.

DOCUMENT CONTROL, VERSIONING & CONTENT MAP

This BRD establishes a controlled baseline for a 40-page scientific requirements document. The scope moves from cosmology to galaxies, then to the Milky Way, stars and compact objects, the Solar System and Earth, exoplanets and life, exploration systems, scientific data, hazards, and future research. Each page is intended to be independently useful while remaining part of one logical chain.

Document ownership should remain with the designated scientific program owner or academic project lead. Changes that alter scientific scope, requirement meaning, evidence level, mission status, or acceptance criteria should receive review before incorporation. Editorial corrections may be applied without changing the version purpose, while substantive changes require a version increment and a documented review note.

Intended audience includes astronomy learners, business analysts working with scientific programs, research coordinators, mission-planning teams, data analysts, educators, technology teams and reviewers assessing the completeness of a space-science knowledge framework.

Control Item	Baseline
Document ID	BRD-SPACE-001
Version	1.0
Status	Baseline Draft / Review Ready
Owner	Space Science Knowledge Initiative
Primary audience	Research, education, analysis and project teams
Change control	Impact review → scientific review → approval → version update
Evidence rule	Established observations separated from hypotheses and open questions

- Page map: 1 Cover; 2 Control; 3 Executive Summary; 4 Universe; 5 Origin; 6 Timeline; 7 Cosmic Scale; 8 Galaxies; 9–11 Milky Way; 12–18 Stellar science; 19–24 cosmology and structure; 25–28 Solar System, Earth and life; 29–35 exploration and technology; 36–40 research requirements, data, future, risk and sign-off.
- Review history: scientific accuracy, terminology, requirement traceability, page completeness and reference integrity should be reviewed separately.
- Approval history: final approval should be recorded only after the named scientific reviewer confirms that factual claims and mission-status statements are appropriately sourced.

EXECUTIVE SUMMARY

The Universe is the total physical system accessible to scientific description, encompassing spacetime, matter, radiation and the large-scale structures that have developed over cosmic history. The observable Universe is the region from which information has had time to reach Earth. The distinction is operationally important: observation is limited by the finite speed of light and cosmic history, while the total Universe may extend beyond what can currently be observed.

This BRD converts a broad astronomy topic into a structured requirements framework. Its central requirement is coherence: cosmological history must connect logically to the formation of galaxies; galaxy evolution must connect to the Milky Way; stellar evolution must explain the origin of many elements and compact remnants; planetary science must place Earth in context; and exploration requirements must reflect what observations and technology can realistically deliver.

The major scientific domains include expansion and cosmology, the cosmic microwave background, dark matter and dark energy, galaxy formation, stellar birth and death, black holes and neutron stars, interstellar matter, planetary systems, exoplanets, astrobiology, gravitational waves and multi-wavelength astronomy. The document also treats research as an operational process involving observation, calibration, modeling, validation, archiving, reproducibility and communication.

Several questions remain open. The nature of dark matter and dark energy is unresolved; the earliest stars and galaxies are still being reconstructed from limited observations; the origin of supermassive black holes remains an active research area; the prevalence of habitable environments is uncertain; and the physical conditions necessary for life beyond Earth have not yet been shown to produce confirmed extraterrestrial life.

Theme	Core requirement	Outcome
Cosmology	Connect observations to physical models	Coherent cosmic history
Galactic science	Map structure and evolution	Milky Way and galaxy context
Stellar science	Trace mass-dependent evolution	Lifecycle pathways
Planetary science	Compare worlds and systems	Context for habitability
Exploration	Define science-driven mission needs	Prioritized research roadmap

BRD intent Use the document as a baseline for planning, review, educational delivery or scientific-program analysis—not as a claim that all open questions have been solved.

INTRODUCTION TO THE UNIVERSE

In modern cosmology, the Universe is modeled as an evolving spacetime containing matter, radiation and energy. General relativity supplies the principal large-scale framework for gravity, while quantum theory describes microscopic interactions. A complete theory that unifies these descriptions is not yet available, so the earliest moments of cosmic history remain partly theoretical.

The observable Universe is defined by the region from which electromagnetic signals or other information could have reached us since the hot early state. It is therefore not simply a sphere with Earth at a physical center; it is an observational boundary produced by cosmic age, expansion and signal travel time. Looking farther away also means looking farther into the past.

On the largest scales, observations show a Universe containing galaxies arranged in groups, clusters, filaments and voids. At smaller scales, matter forms stars, planets, dust clouds and biological environments. The same physical principles operate across these scales, but different approximations and instruments are needed. Astronomy therefore depends on a hierarchy of models rather than one universal measurement technique.

Current observations support a hot, expanding early Universe, the existence of the cosmic microwave background, large-scale structure growth, and gravitational effects attributable to unseen mass. They also indicate an accelerating late-time expansion. The composition is commonly summarized in approximate fractions of ordinary matter, dark matter and dark energy, but those fractions are model-dependent cosmological parameters rather than directly visible substances.

Scale / entity	Representative description	Primary scientific method
Earth	Planetary environment	In situ + remote sensing
Solar System	Planetary and small-body system	Orbital dynamics + spacecraft
Milky Way	Galactic ecosystem	Optical, infrared, radio, X-ray surveys
Galaxy groups/clusters	Bound large-scale systems	Imaging + spectroscopy + lensing
Cosmic web	Largest mapped structure	Redshift surveys + simulations
Observable Universe	Finite observational domain	Multi-messenger cosmology

Figure 1 – Suggested conceptual diagram Show nested scales from Earth to Solar System, Milky Way, galaxy groups, cosmic web and observable horizon, with arrows indicating increasing distance and look-back time.

ORIGIN OF THE UNIVERSE

The standard Big Bang model does not describe an explosion into pre-existing empty space. It describes an early hot, dense state followed by the expansion and cooling of spacetime. As the temperature fell, physical processes changed: elementary particles could persist, light nuclei formed, neutral atoms eventually appeared, and later gravitational collapse built stars and galaxies.

Inflation is a proposed very early period of accelerated expansion that can account for the large-scale uniformity and near-flatness of the observable cosmos while providing a mechanism for generating primordial fluctuations. Inflation is a leading framework in modern cosmology, but the physical field responsible, its exact duration, and the transition to the hot early phase are not directly established by a complete experimental theory.

Primordial nucleosynthesis provides a major quantitative test. Within minutes of the hot beginning, nuclear reactions produced much of the Universe's hydrogen and helium and small quantities of other light nuclei. Later, around hundreds of thousands of years after the beginning, electrons combined with nuclei to form neutral atoms, allowing photons to travel more freely and producing the relic radiation observed today as the cosmic microwave background.

Evidence for the hot early Universe comes from multiple independent observations: cosmic expansion, the cosmic microwave background, measured light-element abundances, and the growth of large-scale structure. These observations strongly support the broad cosmological model. They do not, by themselves, establish every proposed description of the earliest instant.

Claim	Evidence status	Interpretation
Universe expanded from a hotter, denser state	Strongly supported	Core Big Bang framework
Light-element abundances reflect early nuclear reactions	Strongly supported	Primordial nucleosynthesis
CMB is relic radiation from early transparent Universe	Strongly supported	Last-scattering evidence
Inflation occurred exactly as a specific model predicts	Open/model-dependent	Active research
Physics at the first instant is fully known	Not established	Requires quantum-gravity insight

Evidence discipline The document distinguishes measured phenomena from explanatory models. A successful model can be strongly supported without every underlying physical mechanism being directly observed.

COSMIC TIMELINE

Cosmic chronology is a sequence of physical regimes rather than a simple calendar. The earliest interval is the least certain because established theories cannot yet provide a complete description of quantum gravity. Later milestones become progressively more observationally constrained as nuclear physics, recombination physics, relic radiation and structure formation become testable.

After the earliest hot state, rapid expansion and cooling permitted particle processes to settle into conditions described by the Standard Model and cosmological physics. Within minutes, primordial nucleosynthesis created light nuclei. Hundreds of thousands of years later, recombination produced neutral atoms and the Universe became transparent to the radiation that is now observed as the cosmic microwave background.

The following period is called the Cosmic Dark Ages because there were no stars producing the familiar starlight yet. Gravity amplified small density variations. The first stars transformed the radiation and chemistry of their surroundings, and later generations of stars enriched gas with heavier elements. Galaxies assembled through gravitational growth, gas inflow, star formation, feedback and mergers.

Modern cosmic history is therefore a feedback system. Stars alter their environments; galaxies regulate how gas becomes stars; black holes can inject energy into surrounding material; and dark matter provides a gravitational framework. Observations of increasingly distant galaxies let astronomers reconstruct different stages of that history.

Milestone	Approximate era	Scientific significance
Planck-scale epoch	Very early; physics unresolved	Quantum gravity regime
Inflation / reheating	Very early; model-dependent	Primordial conditions
Nucleosynthesis	First minutes	Light nuclei
Recombination	~380,000 years	Atoms + CMB release
Dark Ages	After recombination	Pre-stellar interval
First stars	Hundreds of millions of years	First stellar radiation
First galaxies	Early cosmic history	Hierarchical structure growth
Present epoch	~13.8 billion years after hot beginning	Accelerating expansion + mature structure

Figure 2 – Suggested timeline Use a horizontal logarithmic time axis because early events occupy a tiny fraction of cosmic history compared with the age of galaxies and stars.

SCALE OF THE COSMOS

Astronomical distances require units that match the problem. Kilometres are practical for planetary distances and spacecraft navigation, the astronomical unit (AU) is convenient for Solar System scales, light-years describe the distance light travels in one year, and parsecs arise naturally from stellar parallax. Larger structures are commonly expressed in kiloparsecs, megaparsecs and gigaparsecs.

An important analytical distinction is between distance and look-back time. A source seen at great distance is observed at an earlier stage of its history because its light required time to travel to us. Cosmic expansion makes the relation between present-day distance, emitted-light distance and redshift more subtle than simply multiplying a time by the speed of light.

Scale is also a requirement for instrument design. A telescope resolving a planetary disk, a survey mapping millions of galaxies, and a detector measuring a gravitational-wave strain operate on different angular, temporal and sensitivity regimes. Requirements therefore need to specify both the physical scale of the target and the measurement scale of the instrument.

Reference	Approximate scale	Useful unit
Earth–Moon	3.84×10^5 km	km
Earth–Sun	1 AU	AU
Nearest stellar systems	Several light-years	ly / pc
Milky Way disk	$\sim 10^5$ ly order	kpc / ly
Nearby galaxy distances	Millions of ly	Mpc
Large cosmological surveys	Billions of ly	Mpc / Gpc
Observable cosmos	Tens of billions of ly in present comoving scale	Gpc

- 1 AU is the mean Earth–Sun distance and is defined exactly as 149,597,870,700 m.
- A parsec is approximately 3.26 light-years and is tied to the geometric parallax of an object.
- Cosmological distances should specify the adopted distance definition when precision matters: luminosity distance, angular-diameter distance and comoving distance are not interchangeable.
- Figure 3 – Suggested logarithmic scale ladder: Earth → Moon → AU → stellar neighborhood → Milky Way → local galaxy environment → cosmic web → observable Universe.

GALAXIES

A galaxy is a gravitationally bound system containing stars, stellar remnants, gas, dust and dark matter. Galaxies are not static collections. Their shapes, star-formation rates, gas reservoirs and central black holes can change through internal processes and interactions with their environment. Classification provides a useful observational language, while evolution studies seek the physical causes behind the observed diversity.

Spiral galaxies generally possess rotating disks, central bulges and structured arms associated with density patterns and star-forming regions. Elliptical galaxies are dominated by older stellar populations and often have little cold gas available for current star formation. Irregular galaxies lack a simple regular morphology and can be strongly affected by interactions, feedback or small mass.

Dwarf galaxies are particularly important because they are numerous and can preserve evidence about early structure formation and feedback. Groups and clusters extend the hierarchy beyond individual galaxies. In clusters, gravitational interactions, ram-pressure stripping, mergers and hot intracluster gas can change how member galaxies evolve.

Galaxy mergers can trigger star formation, redistribute stars and gas, and influence central black-hole activity. Over cosmic time, the observed galaxy population emerges from the combined effects of dark-matter assembly, gas cooling, star formation, stellar feedback, black-hole feedback and environmental processing.

Class	Typical properties	Key research question
Spiral	Disk, rotation, arms, ongoing star formation	How do disks maintain and transform structure?
Elliptical	Older populations, little cold gas	Why did star formation cease?
Irregular	Disturbed/asymmetric morphology	What processes caused the disturbance?
Dwarf	Low mass, often gas-sensitive	How do small halos form and retain baryons?
Cluster	Many galaxies + hot intracluster medium	How does environment regulate galaxies?

Figure 4 – Suggested classification plate Present representative spiral, elliptical, irregular and dwarf morphologies alongside notes on gas content, stellar age and environment.

THE MILKY WAY

The Milky Way is the barred spiral galaxy that contains the Solar System. It is a large, rotating stellar system with a disk, central bulge/bar region, extended stellar halo and a much larger dark-matter halo inferred from gravitational dynamics. Its structure is reconstructed from observations of stars, gas and dust across many wavelengths because Earth is embedded inside the disk rather than viewing the galaxy from outside.

The Galactic disk contains most of the Milky Way's young stars, gas and dust. Spiral structure is traced by combinations of star-forming regions, molecular clouds, young stellar populations and gas kinematics. The central bulge and bar host dense stellar populations and connect to the Galactic center. The halo contains old stars and globular clusters and extends far beyond the bright disk.

The Solar System lies in the Orion Arm or Local Spur region, at a distance of roughly 8 kiloparsecs from the Galactic center. The Sun follows an orbit around the center rather than remaining fixed relative to the spiral pattern. Precise mapping requires astrometry, spectroscopy and models of Galactic dynamics.

The Milky Way is also a historical record. Stellar ages, metallicities and motions provide clues about when different populations formed, how the disk assembled, and how mergers contributed to its present structure. Modern surveys turn millions of stellar measurements into a four-dimensional picture involving position, velocity and chemical composition.

Component	Role in Galactic structure	Primary tracers
Thin disk	Young stars, gas, current star formation	Optical/IR + radio
Thick disk	Older, dynamically hotter stars	Astrometry + spectroscopy
Bulge/bar	Dense central stellar component	Infrared + stellar kinematics
Stellar halo	Old, extended population	Globular clusters + halo stars
Dark halo	Dominant inferred mass component	Rotation + dynamics + lensing

Figure 5 – Suggested Milky Way cross-section Show disk, bulge/bar, halo, spiral features and the Solar System's approximate location without implying that the diagram is to scale.

STRUCTURE OF THE MILKY WAY

Understanding the Milky Way requires separating geometric components that overlap in projection. The thin disk is relatively young and dynamically cold, while the thick disk contains older stars with larger vertical motions. The bulge is a dense central population, and the bar is an elongated structure crossing the inner region. Beyond these components lies a diffuse stellar halo populated by old stars and globular clusters.

The interstellar medium connects these structures physically. Atomic gas, molecular gas, ionized gas and dust occupy different phases and temperatures. Molecular clouds provide the dense reservoirs from which many stars form. Supernovae, stellar winds and radiation return energy and material to the surrounding medium, creating a cycle between stars and gas.

Spiral arms are not rigid material structures in the simplest interpretation. They are patterns in the disk that can organize gas compression and star formation. Their detailed origin and longevity remain active topics, especially when considering the influence of the bar, gravitational interactions and transient density structures.

Galactic archaeology combines chemistry and motion. A star's elemental abundances act as a record of the gas from which it formed, while its orbit provides dynamical history. Large stellar surveys therefore support a form of reconstruction in which millions of individual measurements are combined into population-level evidence.

Component	Approximate characteristic	Research use
Thin disk	Flattened, star-forming	Recent Galactic history
Thick disk	More vertically extended	Early disk evolution
Bar/bulge	Dense inner region	Central dynamics
Globular clusters	Old bound stellar systems	Halo history
ISM	Gas + dust in multiple phases	Star formation and enrichment

Figure 6 – Conceptual cross-section Illustrate vertical structure from the thin disk through the thick disk into the stellar halo, with a separate inset for the barred central region.